CSA07 — Computer Networks

Assessment Tool 2 — Case Study (CO3)

Q1. RIP — Chennai HQ Campus

a) Understanding of the Case

Between 5 and 9 February 2024, the Delivery IT team requested a chain of temporary VLAN gateways at Chennai HQ to onboard a new client (CHG-0221), and a new Delivery subnet was provisioned behind that chain five days later (CHG-0224). Both changes were logged as Medium risk and approved by the Network Supervisor, but neither change record shows a hop-count or reachability check against the existing RIP v2 design running across the Finance, HR and Delivery subnets.

On 18 March 2024 at 09:42 IST, Delivery raised the first ticket: the newly onboarded application subnet was unreachable. NOC engineers confirmed at 10:15 IST that the subnet simply did not appear in the HQ core switch's routing table — not that it was flapping or slow, but that RIP had never installed a route to it at all. The trigger was the VLAN gateway chain added in CHG-0221: each gateway in the chain adds one router hop, and by the time traffic reached the Delivery subnet it had crossed 16 hops from the core switch.

This went undetected for over four months (Feb 9 → Mar 18) because RIP v2 does not raise any error, log entry, or SNMP trap when it excludes a route for exceeding the hop limit — it simply omits the prefix from its updates. NOC monitoring at NimbusEdge (per the RCA's contributing-factors section) was built to alert on link-down and interface-down events, not on the absence of a route in the table. The fault was purely logical/administrative, so it stayed invisible until a client actually tried to use the subnet and generated a support ticket. The emergency fix (CHG-0256) was a manual static route inserted during the incident as a stop-gap, not a redesign.

b) Application of Concepts

RIP v2 is a distance-vector protocol whose only metric is hop count, with a defined maximum of 15 usable hops for a route to be considered reachable. Applying that rule to the Delivery chain step by step:

- Hop 0: Core switch — origin of the route computation.
- Hops 1–14: Each VLAN gateway in the temporary chain increments the metric by 1 as it re-advertises the subnet toward the core. Up through hop 14 the route is still valid and gets propagated normally.
- Hop 15: The last gateway RIP will still advertise as reachable. A route with metric 15 is still installed — this is the boundary of RIP's usable range.
- Hop 16: One VLAN gateway too many — the effective distance from the core switch to the Delivery subnet is 16 hops. RFC 2453 defines any route with metric 16 as 'infinity', i.e. unreachable. When the local router computes a next-hop metric of 16, it does not install the route in the routing table, and it advertises it to neighbours as unreachable (or simply withholds it, depending on implementation) rather than as a usable path.

This is exactly what the NOC observed: not a flapping or partially-working subnet, but a subnet completely absent from the routing table, with no distinguishing error code — because to RIP, metric 16 and 'this network does not exist' look identical.

c) Analysis

No automatic alert fired because a distance-vector protocol like RIP treats hop-count exhaustion as a normal, silent outcome of route computation, not as a fault condition. RIP was designed on the assumption that a well-run network stays within roughly 15 routers' diameter; going past that is not something the protocol flags to the administrator — it is simply the mechanism RIP uses to prevent count-to-infinity routing loops. In other words, the protocol has no vocabulary for 'this route is missing because it's too far away' versus 'this route was never configured' — both look like 'not in the table'.

This is a structural limitation, not a one-off human mistake in the traditional sense. The human error (stacking multiple VLAN gateways without checking cumulative hop count) was the trigger, but the reason it turned into a four-month-long undetected outage risk is that RIP itself provides no telemetry for administrators to catch the condition proactively. This lines up directly with the RCA's stated organisational gap: monitoring tools alerted on physical link-down events but 'not on silent routing anomalies (RIP unreachable routes, OSPF adjacency failures).' The RIP hop-count ceiling and the monitoring gap compounded each other — the protocol hid the problem, and the monitoring stack wasn't built to look for that specific kind of silence.

d) Solution Design

Recommended fix — redesign the addressing path rather than patch around RIP:

- Eliminate the VLAN-gateway daisy-chain: collapse the temporary onboarding chain into a routed VLAN structure with the Delivery subnet attached directly (1–2 hops) to the Chennai HQ core switch/router, using a Layer-3 switch with sub-interfaces or SVIs instead of chained Layer-2/3 gateways.
- Apply a documented VLSM plan for Chennai HQ campus (Finance/HR/Delivery) so that any future subnet addition has a pre-approved attachment point within 2–3 hops of the core — with the CHG process requiring a hop-count check before approval.
- As a protocol-level safeguard, redistribute the Chennai HQ campus into the OSPF backbone as a stub/NSSA area (the migration already underway per CHG-0233 onward) rather than leaving it on pure RIP long-term — this removes the 15-hop ceiling entirely for future growth.

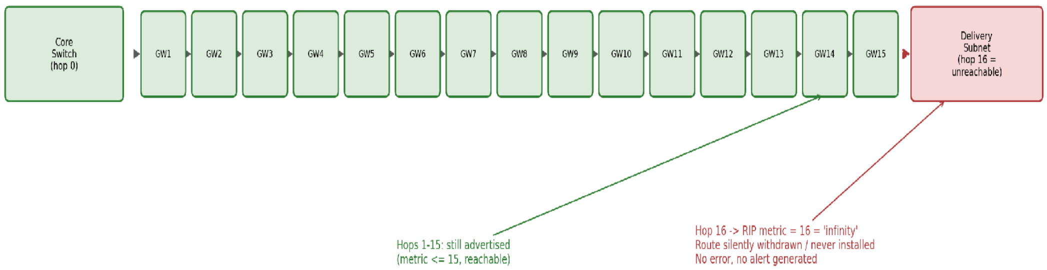
This is feasible for NimbusEdge specifically because the Chennai HQ campus is described in the topology notes as 'small, stable topology' — a flatter VLAN/subnet design does not require new hardware, only reconfiguration, and it aligns with the migration program NimbusEdge has already committed to ahead of the Q3 2025 onboarding surge.

RIP itself was a reasonable original choice for this segment: it is simple to configure and administer, has low CPU/memory overhead suited to a small number of routers, and campus networks of this size rarely approach anything close to 15 hops under normal design. The failure here was not that RIP is a bad protocol for a small stable campus — it is that an unplanned, temporary change (the VLAN gateway chain) was allowed to violate the assumption RIP depends on, without any change-control check for that specific risk.

e) Clarity — Summary

Problem statement	Technical reasoning	Recommendation
A chained VLAN-gateway onboarding change pushed the new Delivery subnet to 16 hops from the Chennai HQ core, and RIP v2 silently excluded it (metric 16 = unreachable) with no alert, going unnoticed for ~4 months until client tickets forced discovery on 18 Mar 2024.	RIP v2's hop-count metric caps usable routes at 15; any route computed at 16 hops is treated as infinite/unreachable per RFC 2453 and is withheld from the routing table without an error condition, matching the NOC's observation of a route missing from the table.	Collapse the VLAN-gateway chain into a direct 1-2 hop attachment for the Delivery subnet, enforce a hop-count check in change control, and migrate Chennai HQ campus routing under the OSPF backbone (stub/NSSA) as part of the existing migration program.

Fig. 2 RIP v2 Hop-Count Exhaustion — Chennai HQ Delivery Subnet



Q2. IGRP — Bengaluru–Pune–Hyderabad Regional Link

a) Understanding of the Case

The Bengaluru–Pune–Hyderabad regional interconnect runs IGRP, inherited from a 2023 acquisition. Per the NOC Change Log, an ISP-side fault on the Bengaluru–Pune fiber segment was first flagged as intermittent on 14 February 2024 (CHG-0229, Low risk, 'Monitoring') and was confirmed as an ongoing, worsening fault on 15 March 2024 (CHG-0252, Medium risk, still 'Open' at the time of the incident). This means the degradation was a known, tracked condition for over a month before it caused a SEV-1 outage.

During the 18 March incident, at 11:05 IST the Bengaluru support desk reported severe application latency (600ms+) on traffic routed via the direct Bengaluru–Pune link, while Hyderabad — reachable via a separate, healthier path — reported normal performance at that point. The topology diagram shows the direct Bengaluru–Pune link is only 2 Mbps, versus an 8 Mbps Pune–Hyderabad leg, so once the fiber fault reduced the effective bandwidth and increased delay on the already-thin 2 Mbps link, the healthier Pune–Hyderabad–Bengaluru alternate should have become clearly preferable. Instead, IGRP kept traffic on the degraded direct path, and business impact was severe, sustained latency for Bengaluru-based client applications for hours until engineers manually adjusted route preference at 18:10 IST (CHG-0257).

b) Application of Concepts

IGRP computes a composite metric primarily from bandwidth and delay (Cisco's default K-values use K1 = K3 = 1, K2 = K4 = K5 = 0), simplified as:

Metric = (10,000,000 / BWmin,Kbps) + (Sum of segment delays, in tens of microseconds)

Applying this to the two candidate paths (illustrative figures consistent with the case data — the topology diagram gives link bandwidths; per-segment delay values are not published in the source documents, so representative delay figures are shown to demonstrate the calculation method expected of engineers reviewing this incident):

Path	Bottleneck BW	BW component (10^7/BW)	Delay component (illustrative)	Composite metric
Direct: Bengaluru → Pune (degraded)	~2 Mbps nominal, effectively lower under fault (assume ~500 Kbps)	10,000,000 / 500 = 20,000	Elevated (assume 4,000)	≈ 24,000 (high = poor)
Alternate: Bengaluru → Hyderabad → Pune	8 Mbps legs (assume 8,000 Kbps bottleneck)	10,000,000 / 8,000 = 1,250	Normal (assume 1,000)	≈ 2,250 (low = preferred)

Because IGRP prefers the path with the lower composite metric, the alternate Pune–Hyderabad–Bengaluru route — with its higher bandwidth and lower delay once the direct link degraded — should have been recalculated as clearly preferred and installed as the active route well before the incident escalated, without requiring manual intervention.

c) Analysis

It is important to separate two distinct kinds of IGRP problem: a metric-calculation issue (IGRP computing the wrong preference given the inputs it has) versus a convergence-speed issue (IGRP eventually computing the right preference, but too slowly relative to the real-world change). The RCA explicitly attributes the fault to the latter: 'IGRP composite metric recalculation and convergence lagged the actual link degradation' — the direction of the metric calculation was not wrong, it simply did not update fast enough.

This incident is a convergence-speed issue. IGRP is a distance-vector protocol that relies on periodic (not event-triggered) updates, holddown timers, and split-horizon/route-poisoning behaviour to avoid loops — mechanisms that inherently trade convergence speed for stability. When the Bengaluru–Pune link degraded gradually rather than failing outright, IGRP's periodic recalculation cycle and its conservatism against flapping meant the composite metric for the direct path did not cross over the alternate path's metric quickly enough, so traffic stayed on the degraded link during the highest-impact window (approximately 11:05–18:10 IST).

d) Solution Design

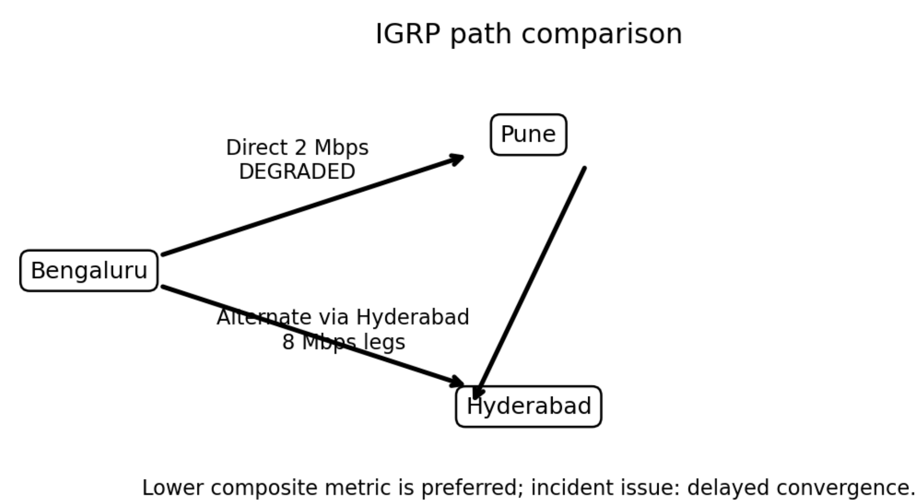
Two options, in order of effort:

- Short-term (tuning): Reduce IGRP update/holddown timers on this regional segment, and manually weight the composite metric (e.g., via offset-lists or static bandwidth/delay statements) to reflect the link's known unreliability, so the alternate path is preferred faster whenever bandwidth or delay crosses a defined threshold.
- Recommended (protocol migration): Extend the OSPF multi-area rollout already underway for the core backbone to cover the Bengaluru–Pune–Hyderabad regional link as well, replacing IGRP. OSPF is link-state and uses event-triggered updates (not periodic polling), so a bandwidth/delay change on an interface is reflected in the link-state database and recomputed via SPF almost immediately, rather than waiting on IGRP's slower distance-vector convergence cycle.

This is justified at NimbusEdge's regional scale: the company is already investing engineering effort in an OSPF migration for the core backbone (CHG-0233 onward), so extending the same design to the regional link reduces the number of routing protocols in production (easier to operate and troubleshoot) rather than tuning a legacy protocol inherited from an acquisition.

IGRP's original advantage over RIP for this segment was its richer composite metric (bandwidth and delay, versus RIP's flat hop count) and its higher hop-count ceiling (up to 255), which suited a regional interconnect spanning several offices with varying link speeds. That advantage failed under real conditions because IGRP's metric richness does not, by itself, guarantee fast reaction to a changing metric — the protocol's underlying distance-vector convergence mechanics (periodic updates, holddown) still governed how quickly that better metric was acted on.

Technical Visual Aid — IGRP Path Comparison

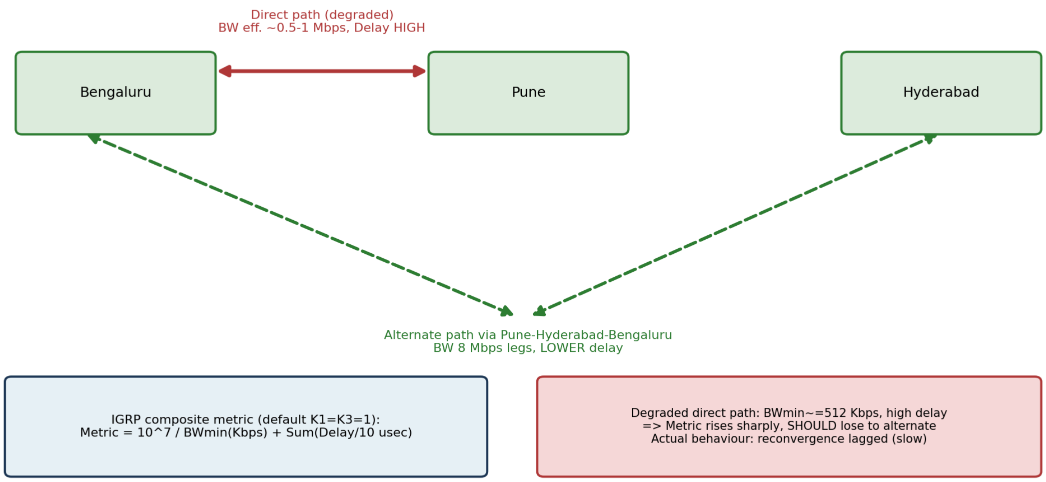


Technical clarification: the incident is best described as a convergence-speed problem. IGRP uses periodic routing updates and additional mechanisms such as triggered/flash updates; its distance-vector convergence behaviour and timers can still delay practical reaction to gradual link degradation.

e) Clarity — Summary

Metric calculation shown above . Root issue: convergence-speed, not calculation-direction (Section c). Recommendation: tune IGRP timers short-term; migrate the regional link into the OSPF multi-area design already in progress for the backbone.

Fig. 3 IGRP Composite Metric — Degraded Direct Path vs. Healthier Alternate



Q3. OSPF — Core Backbone Migration

a) Understanding of the Case

NimbusEdge is migrating its backbone from distance-vector routing to a hierarchical multi-area OSPF design: Area 0 at the Chennai backbone router, with Bengaluru, Pune and Hyderabad as Areas 1, 2 and 3 respectively (per the topology diagram). The NOC Change Log shows this rolled out in phases: CHG-0233 (Chennai Area 0, 20 Feb), CHG-0238 (Bengaluru Area 1, 27 Feb, adjacency verified), CHG-0241 (Pune Area 2, 2 Mar, adjacency verified) and finally CHG-0245 (Hyderabad Area 3, 8 Mar) — the last of which is explicitly flagged in the change log as 'Implemented – Error Introduced': an incorrect Area ID was entered during that configuration.

Unlike the two prior phases, the Hyderabad change record does not show adjacency verification. Because the Area ID mismatch does not bring the physical interface down, the Hyderabad router's link stayed up and passed standard link-state monitoring, while the OSPF neighbor relationship with the Chennai Area 0 backbone router never formed. This created a routing black hole for all Hyderabad-bound traffic that was invisible to interface/link-based monitoring for ten days (8–18 March), and was only surfaced during the incident when, at 12:30 IST, engineers troubleshooting the regional link discovered — via traceroute, not via an alert — that the Hyderabad router had no adjacency with the Chennai backbone router at all, and traffic was being silently dropped.

b) Application of Concepts

OSPF requires several parameters to match exactly between two routers before a neighbor relationship can progress to FULL adjacency: Area ID, Hello and Dead interval timers, authentication type/credentials, and (for broadcast networks) matching subnet mask on the shared segment. If any one of these mismatches, the routers may still exchange Hello packets at Layer 2/3 (so the physical/logical link tests as 'up'), but OSPF will reject the neighbor and adjacency will not form.

In this incident, the violated requirement was Area ID: the Hyderabad router was configured with an Area ID inconsistent with the Area 3 assignment intended for it during the phased cutover (CHG-0245). Because a router's interface Area ID must match the Area ID configured on the neighbouring interface across the link, this single-field transcription error was sufficient on its own to block adjacency formation — Hello/Dead timers, authentication and subnet mask can all be correct and it still fails, because OSPF treats Area ID as a hard prerequisite for accepting a neighbor into the same link-state domain, not a negotiable or auto-correcting parameter.

c) Analysis

This fault was harder to detect than a physical link failure because the symptom and the cause live at different layers. A physical link failure (interface down) is exactly what NimbusEdge's monitoring was built to catch — the RCA states monitoring alerted on link-down events. An Area ID mismatch, by contrast, leaves the physical/data-link layer completely healthy; the failure is purely in the OSPF control-plane's neighbor state machine, which never progresses past the Init/ExStart stage for a mismatched Area ID. Nothing in a link-state or interface-status dashboard would show this as a problem — an engineer has to specifically check OSPF neighbor tables or run a traceroute (as ultimately happened) to notice the routing black hole.

The organisational gap the RCA identifies compounds this: there was 'no formal change-freeze or validation window... for the concurrent OSPF migration and client onboarding activity,' and — critically — the Hyderabad change record lacks the adjacency verification note that both prior phase changes (Bengaluru, Pune) explicitly include. A peer-review or mandatory post-change verification step, applied consistently across all four phases, would have caught the mismatch on 8 March instead of 18 March. A single transcription error is a normal, expected class of human mistake; what let it persist for ten days and cascade into a SEV-1 outage was the absence of a process step to independently verify adjacency after every phase of a high-risk (per the change log's own 'High' risk rating) migration.

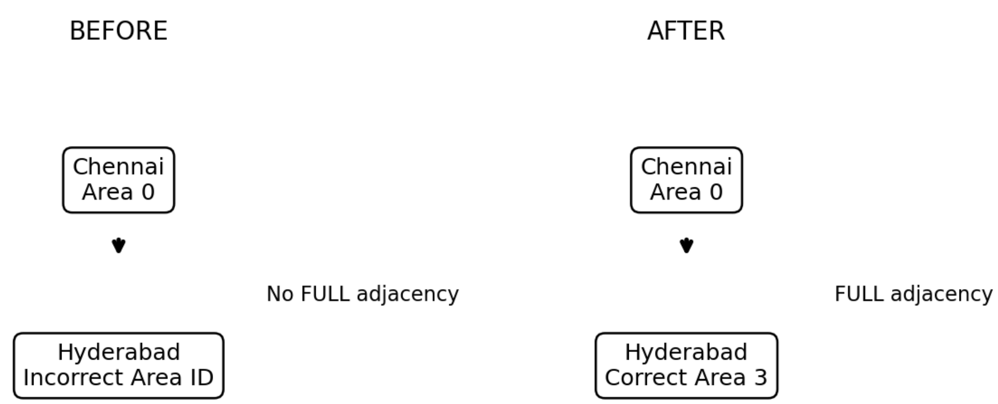
d) Solution Design

Technical fix: correct the Hyderabad router's Area ID to Area 3 as intended (this is exactly what the emergency change CHG-0258 did on 18 March), and, as standard practice for every future OSPF phase, run 'show ip ospf neighbor' (or equivalent) immediately after each area cutover to confirm the neighbor state reaches FULL — not just that the interface reports up.

Process-level safeguard: require a second engineer to peer-review Area ID (and Hello/Dead/authentication) values against the design document before any OSPF phase change is marked implemented, and add an explicit 'Adjacency Verified: Yes/No' field to the change log template so a missing verification is visible at a glance — the way it already was, retroactively, in this incident's log.

NimbusEdge is migrating to multi-area OSPF because it directly addresses the weaknesses exposed by RIP and IGRP elsewhere in this same incident: OSPF is link-state, so it converges quickly and accurately on real topology changes (rather than IGRP's slow periodic distance-vector convergence); it hierarchically scales through areas, so Area 0 carries only summarized inter-area routes rather than every campus/regional route (unlike RIP's flat, hop-limited design); and its link-state database gives every router a complete, consistent view of its area's topology, avoiding the kind of silent, unadvertised gaps that hop-count exhaustion produces in RIP. The trade-off, as this incident shows, is that OSPF's stricter adjacency requirements make configuration accuracy more consequential — a benefit in normal operation, but one that demands the process safeguards above.

Technical Visual Aid — OSPF Adjacency Before/After



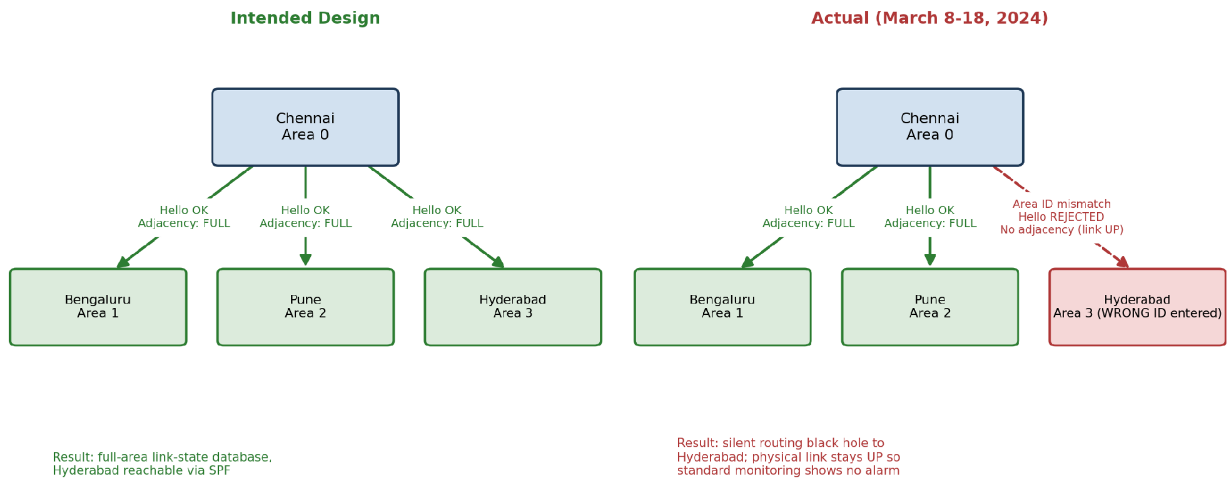
Interface remained UP; the OSPF control-plane adjacency was the key difference.

Before CHG-0258, the physical interface remained up but the incorrect Area ID prevented the intended OSPF neighbor relationship from reaching FULL. After the correction, the adjacency formed and Hyderabad became reachable through SPF-computed routes.

e) Clarity — Before/After

State	Hyderabad Router Area ID	Adjacency with Chennai Area 0	Link/Interface Status	Result
Before (8–18 Mar 2024)	Incorrect (transcription error, CHG-0245)	Never formed	Up (no monitoring alarm)	Silent routing black hole to Hyderabad
After (18 Mar 2024, CHG-0258)	Corrected to Area 3	FULL (verified)	Up	Hyderabad reachable via SPF-computed routes

Fig. 4 OSPF Neighbor Adjacency — Area ID Requirement



Q4. BGP — ISP Peering and Failover

a) Understanding of the Case

NimbusEdge maintains BGP sessions with two ISPs for redundant internet access: ISP-A as primary, ISP-B as secondary, designed so that outbound client-facing traffic normally prefers ISP-A and fails over automatically to ISP-B if the ISP-A session is lost. At 14:50 IST on 18 March, ISP-A reported a fiber cut affecting NimbusEdge's leased line. The expected behaviour was an automatic BGP failover to ISP-B; instead, all outbound client-facing traffic stalled — NimbusEdge continued attempting to prefer the now-unreachable ISP-A path rather than shifting to ISP-B.

Restoring service required a senior network architect to manually intervene at 15:40 IST, forcing traffic onto ISP-B; the ISP-B path was confirmed active and partial restoration began at 16:20 IST (formalised as emergency change CHG-0259). The RCA attributes the failed automatic failover to an untested LOCAL_PREF/AS-PATH outbound policy — last reviewed on 11 March (CHG-0249) but explicitly 'not re-tested end-to-end.' Because the fiber cut occurred during business hours and the automatic mechanism did not engage, the outage window extended to up to 9 hours 13 minutes for affected sites, which is what triggered the formal SLA Breach Notice issued to six enterprise clients, referencing 'Routing infrastructure fault — internal (RIP/IGRP/OSPF) and external (BGP/ISP)' as the root cause category and confirming the monthly 99.9% uptime SLA was breached for the affected billing cycle.

b) Application of Concepts

Three BGP path attributes govern outbound (and, in principle, inbound) path selection here:

- LOCAL_PREF — a locally significant attribute used to prefer one exit path over another for outbound traffic; higher LOCAL_PREF wins, and it is evaluated before AS-PATH length in the BGP best-path algorithm.
- AS-PATH — the sequence of ASes a route has traversed; all else equal, a shorter AS-PATH is preferred, and AS-PATH prepending is often used deliberately to make a path look less attractive so a peer avoids it.
- MED (Multi-Exit Discriminator) — a hint to an external neighbour about which of several entry points it should prefer into the AS; it only takes effect when comparing routes from the same neighbouring AS and is evaluated later in the decision process than LOCAL_PREF.

The RCA points specifically at the LOCAL_PREF/AS-PATH policy as the likely fault. Given that BGP evaluates LOCAL_PREF ahead of AS-PATH, the most probable misconfiguration is that ISP-A's route retained a statically high LOCAL_PREF that was never conditioned on session/liveness state, so even though the physical ISP-A session was down, the router's route-selection policy still treated the (now withdrawn or stale) ISP-A path as preferred rather than falling through to ISP-B's lower LOCAL_PREF route — combined with AS-PATH prepending on the ISP-B route (used to keep it as backup-only under normal conditions) that was never re-evaluated for a scenario where ISP-A actually failed. In short: the policy correctly expressed 'prefer ISP-A normally,' but nothing in it was validated to correctly express 'and switch cleanly to ISP-B when ISP-A is unreachable.'

c) Analysis

An untested failover policy is a materially higher risk in BGP than in an interior gateway protocol because BGP failover, in this architecture, is the last line of defence for external reachability — there is no equivalent of an OSPF SPF recomputation or an IGRP alternate path inside the same AS to fall back on if the exit-path policy itself is wrong. Interior protocols like OSPF/IGRP are self-healing within the AS by design (multiple internal paths, automatic recalculation); BGP's failover, by contrast, depends entirely on a policy that a human wrote and that only gets exercised during an actual ISP failure — precisely the condition it is hardest to safely rehearse, and precisely why the RCA's recommendation is a scheduled test rather than 'fix it once.'

The business consequence chain is direct and traceable through the documents: an unvalidated LOCAL_PREF/AS-PATH policy (RCA, Section 3.4) → automatic failover did not trigger when ISP-A's fiber cut occurred (Incident Report, 14:50 IST) → all outbound client-facing traffic stalled until manual intervention 50 minutes later (15:40 IST) and partial restoration only from 16:20 IST → downtime accumulated up to 9 hours 13 minutes across four sites → six enterprise clients experienced degradation or full unavailability → formal SLA Breach Notice issued, contracted 99.9% monthly uptime breached, and service credits triggered for the affected billing cycle. A single untested policy field is therefore directly connected, through a short and well-documented chain, to a contractual and financial consequence.

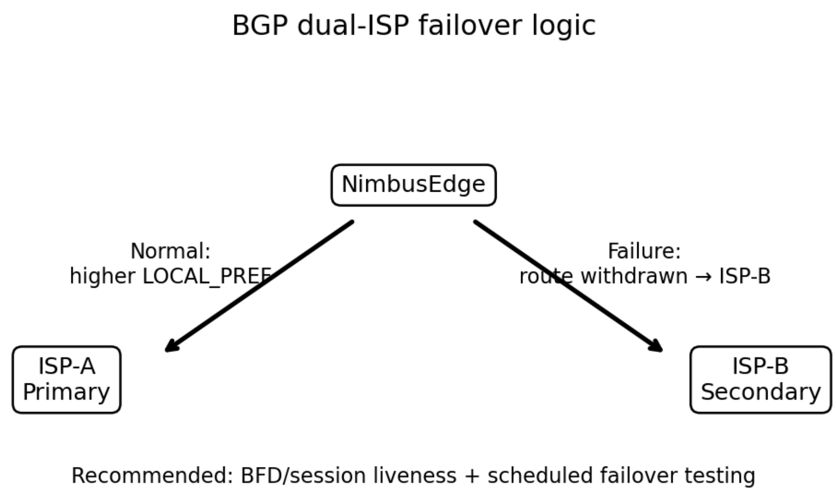
d) Solution Design

Corrected policy: set LOCAL_PREF on the ISP-A route higher than ISP-B's as before (e.g., ISP-A = 200, ISP-B = 100) for normal-state preference, but tie the ISP-A route's presence in the routing table to session/BFD liveness rather than a static preference alone — i.e., ensure the policy is structured so that route withdrawal on BGP session loss (ideally accelerated with BFD for sub-second detection) removes the ISP-A path from consideration entirely, at which point BGP's best-path algorithm has only the ISP-B route left and installs it automatically, with no manual override required.

- Enable BFD on both eBGP sessions to detect the ISP-A fiber-cut condition in seconds rather than relying on default BGP hold timers.
- Establish a quarterly, scheduled maintenance-window failover test (as the RCA itself recommends) that deliberately drops the ISP-A session and confirms traffic shifts to ISP-B within an agreed time, then reverts.
- Document the expected AS-PATH/LOCAL_PREF/MED values for both normal and failover states, and require peer review of any change to this policy, given its High/Critical risk classification in the change log.

BGP — rather than an interior gateway protocol — is the correct and necessary choice for the dual-ISP architecture because BGP is purpose-built for inter-AS routing between independently operated networks (NimbusEdge and its two ISPs are separate autonomous systems); it is the only routing protocol suited to expressing policy-based path selection across administrative boundaries (which path to prefer, how to influence a neighbour's inbound preference via MED, how to make a backup path deliberately less attractive via AS-PATH prepending). Interior protocols like OSPF or IGRP have no concept of autonomous systems or inter-provider policy and cannot be used to peer with an external ISP at all — BGP is the only protocol in this stack designed for that role.

Technical Visual Aid — BGP Failover Logic



The recommended design keeps ISP-A as the normal preferred exit, but makes the preferred route dependent on session/liveness state. When ISP-A fails, its route should be withdrawn and ISP-B should become the best available path automatically.

e) Clarity — Before/After Policy Comparison

State	ISP-A LOCAL_PREF	ISP-B LOCAL_PREF	Failure detection	Failover behaviour
Before (as found, Mar 18)	Static high value, not tied to session state	Static lower value	Default BGP hold timer only; policy untested since Oct 2023	Automatic failover did not trigger; manual override required (CHG-0259)
After (recommended)	High, but route withdrawn immediately on session/BFD loss	Preferred automatically once ISP-A route is withdrawn	BFD-accelerated (sub-second)	Automatic failover to ISP-B within seconds; validated quarterly

Submission Requirement — GitHub

GitHub Repository / Folder: [PASTE YOUR ACTUAL GITHUB LINK HERE]
Before GCR submission, replace the placeholder with your actual GitHub repository/folder link. Recommended contents: the final Word report, Q1–Q4 folders, diagrams, and README.md.